

Digital image processing (DIP)

Lab Projects

CLO-2	<i>Develop and implement a project that applies digital image processing techniques to solve real-world applications</i>		<i>Apply</i>	PLO-4
--------------	--	--	--------------	--------------

Project Objective

The objective of this project is to enable students to:

- Explore recent research in Digital Image Processing.
- Study and analyse research papers related to course topics.
- Implement algorithms and techniques presented in the selected paper.
- Evaluate results and compare findings.
- Write a technical report in IEEE Conference Paper format.

Students are required to select a research paper related to the topics covered in the course and implement the proposed methodology using MATLAB or Python/OpenCV.

Important dates

Activity	Date
Project Topic & Research Paper Selection	14 May 2026
Final Submission Deadline	02 June 2026

Project Requirements

Each student/group must:

1. Select a research paper related to Digital Image Processing.
2. Implement the methodology proposed in the paper.
3. Use datasets/images for experimentation.
4. Analyse results and compare findings.
5. Prepare a complete report in IEEE Conference Style.

6. Present the implementation and results during viva.
7. The group have maximum of 3 students.

Suggested Project Areas & Research Topics

1. Noise Removal and Image Restoration

Related Topics

- Gaussian Noise
- Salt & Pepper Noise
- Adaptive Filtering
- Frequency Domain Filtering

Example Applications

- MRI image denoising
- Satellite image restoration

2. Edge Detection and Sharpening Techniques

Related Topics

- Sobel Operator
- Prewitt Operator
- Laplacian Filters
- Canny Edge Detection

Example Applications

- Object boundary detection
- Road detection systems

3. Image Segmentation Using Thresholding

Related Topics

- Global Thresholding
- Adaptive Thresholding
- Otsu Thresholding

Example Applications

- Tumor segmentation
- Industrial defect detection

4. K-Means Based Image Segmentation

Related Topics

- Clustering
- Region-based Segmentation
- Color-based Segmentation

Example Applications

- Medical image analysis
- Land cover classification

5. Hough Transform Based Detection

Related Topics

- Line Detection
- Circle Detection
- Edge Detection

Example Applications

- Lane detection
- Shape recognition

6. Morphological Image Processing

Related Topics

- Dilation
- Erosion
- Opening
- Closing

Example Applications

- Fingerprint enhancement
- Character extraction

7. Image Recognition Using Thresholding and K-Means

Related Topics

- Pattern Recognition
- Feature Extraction
- Clustering

Example Applications

- Face recognition
- Fruit classification

8. Medical Image Processing

Related Topics

- Segmentation
- Enhancement
- Restoration
- Recognition

Example Applications

- Brain tumor detection
- X-ray enhancement

9. Satellite Image Processing

Related Topics

- Enhancement
- Segmentation
- Restoration

Example Applications

- Land use classification

- Flood detection

10. OCR and Document Image Processing

Related Topics

- Thresholding
- Morphology
- Segmentation

Example Applications

- Character recognition
- Document enhancement

11. Face Detection and Recognition

Related Topics

- Segmentation
- Feature Extraction
- Edge Detection

Example Applications

- Attendance systems
- Security systems

12. Industrial Defect Detection

Related Topics

- Edge Detection
- Morphology
- Thresholding

Example Applications

- Product quality inspection
- Surface crack detection

13. Brain Tumor Segmentation using Hybrid Thresholding and Morphological Operations

Concepts Covered

- Segmentation
- Adaptive Thresholding
- Morphology
- Edge Detection

Possible Dataset

- MRI Brain Images

14. Deepfake Image Detection using Frequency Domain Analysis

Concepts Covered

- Frequency Domain Processing
- Image Enhancement
- Feature Extraction
- Image Recognition

Possible Application

- Cybersecurity & Media Verification

15. License Plate Detection and Character Segmentation System

Concepts Covered

- Edge Detection
- Morphology
- Segmentation
- Thresholding

Possible Application

- Smart Traffic Systems

16. Retinal Blood Vessel Segmentation for Diabetic Retinopathy Analysis

Concepts Covered

- Histogram Processing
- Segmentation
- Morphology
- Image Enhancement

Possible Application

- Medical Diagnosis

17. Satellite Image Based Land Cover Classification using K-Means Clustering

Concepts Covered

- Clustering
- Segmentation
- Spatial Filtering
- Frequency Domain Enhancement

Possible Application

- Remote Sensing

18. Fingerprint Enhancement and Minutiae Extraction System

Concepts Covered

- Morphology
- Sharpening
- Segmentation
- Spatial Filtering

Possible Application

- Biometric Authentication

19. Low-Light Image Enhancement using Adaptive Histogram Equalization

Concepts Covered

- Histogram Equalization
- Spatial Enhancement
- Filtering

- Image Restoration

Possible Application

- Surveillance Systems

20. Underwater Image Enhancement using Frequency and Spatial Filtering

Concepts Covered

- Image Restoration
- Filtering
- Histogram Processing
- Frequency Domain

Possible Application

- Marine Exploration

General Instructions

- Plagiarism is strictly prohibited.
- Students must properly cite all references.
- The selected paper should preferably be from:
 - IEEE
 - Springer
 - Elsevier
 - ACM
- Students are encouraged to use real-world datasets.
- The implementation must demonstrate practical application of Digital Image Processing concepts covered in the course.

Project Deliverables

Students must submit:

1. IEEE Format Project Report (PDF)
2. Source Code

3. Research Paper Copy
4. Presentation Slides
5. Output Results/Screenshots
6. Dataset Details
7. Viva and Demonstration

Report Writing Guidelines

The report must follow the IEEE Conference Paper format.

Mandatory Sections

1. Title

- Clear and meaningful project title

2. Abstract

- Summary of problem, methodology, and results

3. Introduction

- Background
- Problem statement
- Motivation
- Objectives

4. Literature Review

- Summary of related research papers
- Comparison of existing techniques

5. Methodology

- Dataset used
- Algorithms implemented
- Mathematical/model description
- Flowchart/system architecture

6. Implementation

- Tools and technologies
- Experimental setup
- Processing pipeline

7. Results and Discussion

- Output images
- Accuracy/comparison tables
- Analysis of findings

8. Conclusion

- Summary of work
 - Achievements
 - Future improvements
-

9. References

- IEEE citation format

Project Evaluation Rubrics (100 Marks)

Criteria	Description	Marks
Problem Understanding	Clear understanding of the selected problem domain and application	5
Research Paper Selection	Selection of relevant and quality research paper	5
Literature Review	Analysis and comparison of existing techniques and related work	10
Methodology Understanding	Understanding and explanation of the proposed algorithm/method	10
Implementation	Correct implementation of the research paper methodology	15
Image Processing Techniques	Proper application of DIP concepts such as filtering, segmentation, enhancement, morphology, etc.	10
Results & Performance Analysis	Experimental results, comparison, accuracy, visualization, and discussion	10
Innovation / Improvement	Additional enhancement or improvement beyond the original paper	5
Documentation & IEEE Report	IEEE formatting, technical writing quality, references, and organization	15

Presentation & Viva	Demonstration, explanation, questioning, and overall presentation	15
---------------------	---	----